

- 1 (a) (i)** Taking downwards as positive,

$$v_y^2 = u_y^2 + 2as_y$$

$$v_y^2 = 0 + 2(9.81)(32)$$

$$v_y \approx 25.06 \approx 25 \text{ m s}^{-1}$$

Note: Energy conservation can be used.

$$\text{(ii) } \sin \theta = \frac{v_y}{v} = \frac{25.06}{34}$$

$$\theta = 47.5^\circ$$

$$\text{(b) Average force} = \frac{mv - mu}{t} = \frac{0.130(2.0 - 34)}{0.95} \approx -4.379 \approx -4.4 \text{ N}$$

Magnitude of average force is 4.4 N.

- (c)** If the stone causes a large splash, part of its kinetic energy is converted into the kinetic energy of the water spray; part into thermal energy due to collision with the water molecules; and part into sound energy.

With less kinetic energy, the stone will thus be able to do less work against the resistive force acting against its motion.

Since work done against resistive force = average resistive force $\times$ distance moved against the force, the stone will be slowed down in a shorter distance.